# Data-Efficient Modeling of Hysteresis and Crosstalk for Inverse Kinematics of Soft Manipulators

***Abstract*—Nonlinearities in soft continuum manipulators, arising from material hysteresis and intersegmental coupling in multi-segment robots, present significant challenges for accurate open-loop inverse kinematics (IK) control. In particular, morphable pneumatic chambers adjust their shape and stiffness with internal pressure, increasing force output but also introducing nonlinearities that complicate control. This paper introduces a sequence-based machine learning approach that is data-efficient, modeling and compensating for both hysteresis and crosstalk in systems with morphable chambers. Through systematic comparison of Long Short-Term Memory (LSTM) and Transformer architectures under data-limited conditions, we demonstrate the effectiveness of sequence-based models in capturing temporal dependencies. We then propose a Recursive Segment-wise Crosstalk Compensation (RSCC) pipeline that decomposes control of multi-segment robots into independent single-segment subproblems, reducing training data requirements to 500 samples per segment. Applied to a two-segment morphable-chamber manipulator, RSCC achieves approximately 11% normalized positional error and outperforms a monolithic multi-segment LSTM baseline trained on 2000 samples within the same workspace, highlighting its potential for precise open-loop control in minimally invasive surgical applications.**

***Index Terms*—Soft Robotic Systems; Hysteresis Modeling; Recursive Machine Learning; Open-Loop Control**

## I. Introduction

SOFT robotic manipulators offer compliance, making them well suited for surgical applications that require safe tissue interaction [1]. Fluidic elastomer actuators (FEAs) are soft manipulators which achieve manipulation through internal pressurization. A soft body integrated with $\geq 3$ chambers in a cylindrical configuration enables omnidirectional bending and elongation within compact environments [2].

Our long-term goal is to develop a hybrid manipulator for endoscopic submucosal dissection (ESD). It consists of two soft segments, each containing three morphable chambers, and a distal tendon-driven segment for precise tissue manipulation [3]. During ESD, the proximal soft segments provide improved triangulations. They often operate outside of the visible field [4], and their passive compliance helps regulate contact forces and reduce the risk of perforation. An overview schematic is shown in Fig. 1.

A key advantage of this morphable FEA design is its high payload-size ratio. Soft surgical manipulators must remain compact to access internal anatomy yet deliver sufficient force for effective tissue manipulation [5]. Morphable FEAs can tune stiffness through pressurization-induced strain, which enhances lateral rigidity without additional mechanisms [6]–[8]. These actuators remain soft when unactuated yet can achieve high payload capacity in compact form [9].

However, this pressurization-based stiffening effect introduces substantial nonlinearities. Large chamber deformations

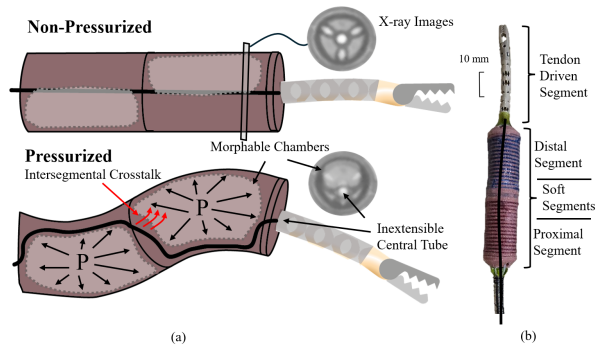


Fig. 1: (a) Illustration of the hybrid soft manipulator in non-pressurized (top) and pressurized (bottom) states. Intersegmental crosstalk (red arrows) results from asymmetric chamber actuation, with X-ray images showing the cross-sectional chamber morphing. (b) Image shows the assembled prototype.

generate pronounced hysteresis due to the hyperelastic material properties [7], [8]. Multi-segment design with a shared central lumen further introduces intersegmental crosstalk [2], [3], [5]. When one segment is actuated, its chamber expansion can deform adjacent chambers directly and via the central lumen, inducing unintended offset in neighboring segments.

To address these nonlinearities, closed-loop control methods are often adopted [3]. However, embedding sensors adds complexity that is undesirable in surgical contexts, motivating research into effective open-loop alternatives. Conversely, open-loop control suffers from cumulative errors due to hysteresis and crosstalk, which degrade accuracy and repeatability unless model parameters are exactly known. While parametric models (e.g., Preisach, Bouc–Wen) can capture elastic hysteresis, they are difficult to calibrate for multi-chamber, coupled morphologies [10]. As demonstrated by Liu et al., chamber coupling creates complex, sequence-dependent dynamics that are challenging to model accurately using analytical or finite element methods [11]. Data-driven techniques such as neural networks have achieved up to 80% accuracy in quasi-static home-to-target tasks [12], but prior work has generally not accounted explicitly for hysteresis and crosstalk.

Sequence-based learning models such as Long Short-Term Memory (LSTM) [13] and Transformer networks [14] have shown strong capability in modeling temporal dependencies in nonlinear systems. In soft surgical manipulators, multi-segment architectures increase redundancy and data requirements, while material degradation limits the amount of data that can be collected [15]. This motivates the development of data-efficient learning frameworks that can accurately model complex hysteresis and intersegmental coupling with limited training data.

To this end, we propose a feedforward inverse kinematics

framework tailored for multi-segment soft manipulators with morphable pneumatic chambers. The main contributions of this work are:

- A modular inverse kinematics pipeline (RSCC) that decomposes multi-segment control into separate single-segment models with learned intersegmental crosstalk compensation, enabling accurate open-loop tracking with substantially fewer training samples than a monolithic multi-segment LSTM on the same robot and workspace.
- A sequence-based learning formulation for soft manipulators with morphable chambers, showing that modest history windows are sufficient to capture nonlinear hysteresis and crosstalk.

## II. State of the Art

### A. Soft Robotics Manipulators for GI Applications

In gastrointestinal (GI) applications, manipulators must have a diameter ($\leq$17 mm) to pass through the colon's narrowest sections [16]. To meet clinical access requirements, FEAs are miniaturized, which often limits their achievable lateral force output [5]. This reduction in size not only impacts force generation but also makes precise bending more challenging. To address these issues, early designs incorporated radial fiber reinforcements to suppress chamber lateral expansion and enhance bending precision [17]. However, this approach reduced the chamber's deformability, which limits payload capacity. Variable-stiffness mechanisms have since been integrated into the designs to mitigate this trade-off [18].

An alternative design relocates reinforcement fibers to the actuator's exterior, allowing internal chambers to strain radially and increase stiffness before and during bending [7], [8]. This morphable-chamber principle enhances both force output and rigidity of fiber-reinforced FEAs with comparable sizes [6]. As summarized in Table I, most sub-15 mm diameter manipulators produce low tip forces, which are insufficient for tissue traction in ESD that requires up to 0.4 N [19]. In contrast, the morphable-chamber designs have demonstrated distal forces of up to 2 N, offering a promising path toward surgically viable soft robotic instruments [6].

TABLE I: Comparison of State-of-the-Art Soft Manipulators d $\leq$ 15 mm

| Designs | Actuation | DOFs | Diameter (mm) | Force(Segment/Device)(N) |
|---|---|---|---|---|
| FR-C [18] | P | 2 | 14.5 | 0.36/N/A |
| FR-C [5] | P | 6 | 9.5 | $<$ 0.1/$<$ 0.03 |
| M-C [3], [12],* | P | 6 | 12 | 2/0.4 |

* This work, P: Pneumatic, FR: Fiber Reinforcement, M: Morphable, C: Chamber

### B. Machine Learning Model for Hysteresis

Nonlinear and hysteretic behaviors of FEAs vary based on specific actuator geometry and material selections. Analytical kinematic models are therefore difficult to generalize across different FEA designs. A prior model-based approach for soft robots with fiber-reinforced chambers [2] achieved a remarkable tip-position control accuracy with a Root Mean Square Error (RMSE) below 4%. However, with the morphable-chamber designs, where hysteresis and crosstalk in multi-segment robots are exaggerated, the applicability of such analytical models is likely limited.

Machine learning (ML) methods with temporal dependencies can capture nonlinear hysteresis and inter-chamber interactions in soft robots without explicit modeling. Several recursive ML architectures have been investigated including recurrent neural networks (RNNs) and LSTMs. For instance, Wu et al. [20] demonstrated that an LSTM trained on 19,635 data points with 50 input time window achieved less than 2% RMSE in tracking arbitrary trajectories of pneumatic artificial muscles. Bai et al. used an LSTM model with a 25-step time window and 3000 task-based samples to learn inverse kinematics for a tendon-driven surgical robot. They achieved mean tip tracking RMSE 4.0–5.8% of the workspace [13]. A Transformer-based segmented learning framework was applied to a cable-driven parallel continuum robot, using 12,000 samples to train feedforward models mapping from task space through joint configuration to actuation space. The model with 1 input time window achieved RMSE 3.5–4.3 % of workspace in trajectory tracking under planar bending [14].

TABLE II: Comparison of State-of-the-Art Inverse Kinematics (IK) ML in Soft Continuum Manipulators (Only Open-Loop Control)

| Robot Designs | Actuation | DOFs/Segments | Model | Gravity | n | %RMSE |
|---|---|---|---|---|---|---|
| PneuNet [20] | P | 1/1 | LSTM | Y | 19635 | 1.95 |
| McKibben Continuum [21] | P | 2/1 | FNN | N | 2000 | 9.13 |
| Tendon-driven Robot [13] | T | 2/1 | LSTM | N | 3000 | 4.9 |
| Tendon-driven Robot [14] | T | 2/1 | T | N | 12000 | 3.7 |
| McKibben Continuum [22] | P | 4/2 | RNN | N | 12000 | 5.4 |
| Soft Continuum [12] | P | 4/2 | FNN | N | 1500 | 21 |
| Soft Continuum * | P | 4/2 | LSTM | Y | 500 E | 11 |

* This work, P: Pneumatic, T: Tendon, n: Training Samples, E: each segment, T: Transformers

As manipulator complexity increases, particularly in multi-segment configurations, much larger datasets are typically required. For instance, Thuruthel et al. [21], [22] reported that controlling a single segment with feedforward neural networks (FNNs) required about 2,000 samples to achieve roughly 9 % RMSE, whereas extending control to two segments with recurrent neural networks required around 12,000 samples to reach approximately 5.4 % RMSE. A comparative summary of these methods is provided in Table II.

## III. Soft Manipulator Characterization

The challenges of open-loop control are clarified by experimentally characterizing two nonlinear behaviors of the soft segments: hysteresis and intersegmental crosstalk. The soft manipulator utilized in this study is a two-segment, pneumatically actuated soft manipulator [3]. Each soft segment is 22 mm long, with an outer diameter of 12 mm, and incorporates three morphable chambers arranged at 120-degree intervals. A non-extensible central tube constrains axial elongation and directs pressurization to induce lateral bending.

### A. Hysteresis in Morphable Chambers

To characterize the system's hysteresis, we performed an experimental study under quasi-static conditions, where velocity effects are considered negligible. First, we recorded pressure-bending angle responses under cyclic actuation for single-chamber, symmetric dual-chamber, and asymmetric dual-chamber configurations. As shown in Fig. 2 (left), all configurations exhibit notable hysteresis, with clear offsets between the loading and unloading paths. We also investigated input coupling in the task space. We commanded the robot to reach an identical target pressure state, $P = (1.6, 1.6, 0)$, via two distinct paths in the input pressure space: (1) inflating chamber 1 ($P = (1.6, 0, 0)$) then chamber 2 ($P = (1.6, 1.6, 0)$), and (2) inflating chamber 2 ($P = (0, 1.6, 0)$) then chamber 1 ($P = (1.6, 1.6, 0)$). Fig. 2 (right) shows that these different

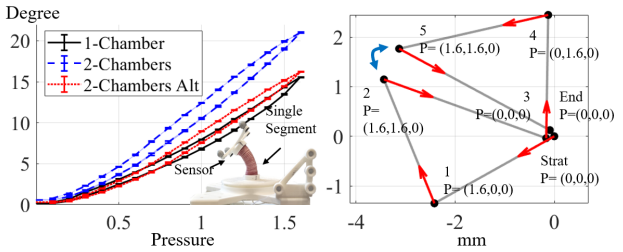


Fig. 2: Experimental characterization of hysteresis. (Left) Pressure-bending angle hysteresis loops for single and dual chamber actuation. Inset: experimental setup with optical tracking markers. (Right) Path-dependent spatial hysteresis in the task space, showing that different pressure input paths (gray lines) to the same target pressure P=(1.6, 1.6, 0) result in different final tip positions (points 2 and 5).

input paths result in two measurably different final tip positions. Results demonstrate that the manipulator's performance depends on both pressure history and the spatial trajectory. To solve for this coupled hysteresis-kinematics problem, a data-driven model capable of implicitly learning these complex, coupled dynamics is an appropriate choice.

### B. Intersegmental Crosstalk

Intersegmental crosstalk is quantified as the unintended motion of one segment caused by the actuation of its neighboring segment. To evaluate this effect, randomized pressure inputs were applied to the proximal segment while the distal segment remained non-actuated. The theoretical neutral position of the distal tip was defined as the point offset by its segment's length along the direction of the proximal tip's orientation. Tip positions and orientations of both segments were recorded using electromagnetic (EM) sensors. The Cartesian crosstalk offsets of the distal tip were computed as the difference between its theoretical and measured positions. A representative deviation of -1.02 mm in $x$ and 0.27 mm in $y$ coordinates was observed when 1.2 bar pressure was applied to chamber 1 of the proximal segment, clearly illustrating the crosstalk effect, as shown in Fig. 3. This displacement offset is used as the system state in the crosstalk prediction model described in Section IV-B2, with experimental validation presented in Section VI-B.

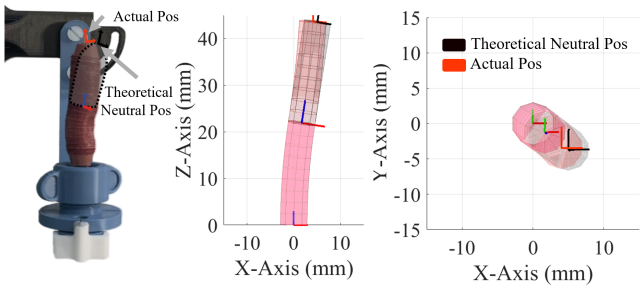


Fig. 3: Illustration of intersegmental crosstalk in a two-segment soft manipulator. Left: Experimental setup with electromagnetic (EM) sensor attached at distal tip. Middle and right: Comparison between actual distal tip position (orange) and theoretical position (black) under proximal actuation, highlighting unintended deflection due to segment coupling.

## IV. Model Formulation & Control Strategy

This section presents our end-to-end control framework for multi-segment soft manipulators. The proposed pipeline, termed Recursive Segment-Wise Crosstalk Compensation (RSCC), integrates optimization-based inverse kinematics with data-driven models to achieve precise open-loop control. We first compute per-segment tip targets via a task-space IK optimization. Each segment's LSTM maps targets to chamber pressures. To handle intersegmental crosstalk, we introduce a crosstalk prediction module that estimates the unintended deformation caused by actuating neighboring segments. Pressures are then iteratively updated using the predicted crosstalk, yielding a scalable pipeline as shown in Fig. 4(a).

### A. Inverse Kinematics for Single-Segment Robot

The IK of each segment is implemented using sequence-based LSTM and Transformer models. Both architectures use historical inputs over a sliding time window, allowing comparison of their effectiveness in capturing segment dynamics and sensitivity to time window length.

Each segment $i$ is actuated by pressurizing the three internal chambers, with control input denoted as $q_i = (p_{i,1}, p_{i,2}, p_{i,3})$. Due to the non-extensible central lumen, segment motion is constrained to omnidirectional bending. Therefore, we define the task space as the projection of the tip position onto the $xy$-plane, $(x_i, y_i)$, and augment it with the bending-plane angle as $\phi_i = \operatorname{atan2}(y_i,\, x_i)$. To avoid angular discontinuities, we embed this angle using its sine and cosine components, forming the state vector $X_i$:

$$X_{i,k} = (x_i, y_i, \cos\phi_i, \sin\phi_i)_k, \tag{1}$$

where $k$ denotes the current discrete time step. To capture temporal dynamics and hysteresis, we define the augmented state vector $U_{i,k}$ as:

$$U_{i,k} = \Big(X_{i,k-1},\ \Delta(X_{i,k})\Big), \text{ where } \Delta(X_{i,k}) = X_{i,k} - X_{i,k-1}. \tag{2}$$

Here, $\Delta X_{i,k}$ explicitly captures the segment's dynamic change, facilitating hysteresis learning. The actuator pressures $\hat{q}_i$ are predicted from a sequence of these input vectors over a time window of length $m$:

$$\hat{q}_{i,k} = \mathcal{L}_i(U_{i,k}, U_{i,k-1}, \ldots, U_{i,k-m}). \tag{3}$$

where, $\mathcal{L}_i\colon \mathbb{R}^{8(m+1)} \rightarrow \mathbb{R}^3$ denotes either the LSTM or Transformer model.

### B. Inverse Kinematics for Multi-Segment Robots

The proposed RSCC pipeline addresses the redundancy in inverse kinematics for multi-segment soft manipulators by mapping a desired end-effector position to appropriate segment-level commands. This is achieved by sequentially decomposing the global IK problem into segment-specific subproblems, enabling scalable and precise control of the entire manipulator.

*1) Redundant IK via Optimization:* We formulate the IK of a two-segment soft manipulator as a constrained optimization problem. The objective is to determine segment-wise configurations that realize a distal tip target $(x_d, y_d)$. Although each segment undergoes 3D bending, we constrain control to the tip's projection in the $xy$-plane due to limited mobility along the $z$-axis imposed by the non-extensible central lumen.

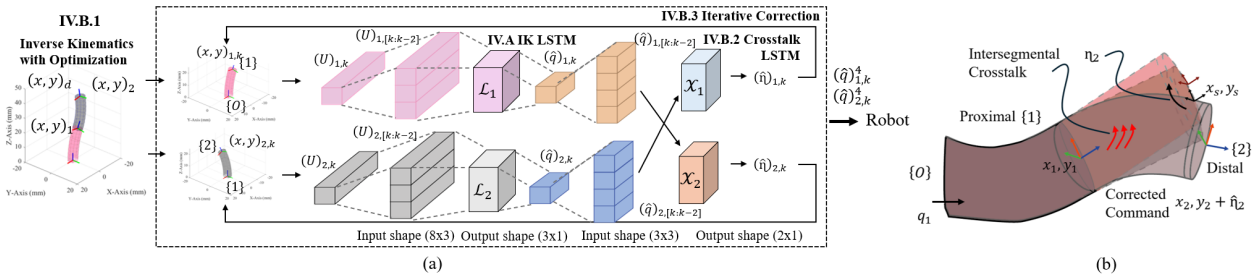


Fig. 4: (a) RSCC pipeline and validation setup. (Left) CC-based IK optimization computes per-segment target tip positions (see Section IV-B1). (Center) IK LSTM predicts pressures (see Section IV-A), crosstalk LSTM estimates disturbances (see Section IV-B2), and an iterative correction loop (see Section IV-B3) refines commands. (b) Free body diagram of intersegmental crosstalk in a two-segment soft manipulator. Pressurizing the proximal segment ($q_1$) perturbs the distal segment via intersegmental crosstalk. This disturbance shifts the neutral tip position $(x_s, y_s)$ of segment 2 by an offset $\eta_2$. This offset is estimated as $\hat{\eta}_{2,k}$ and added to the desired segment position, resulting in the corrected distal task-space position $(x_2, y_2) + \hat{\eta}_2$.

The optimization variables consist of the proximal segment's tip position $(x_1, y_1)$ relative to the base frame and the distal segment's tip position $(x_2, y_2)$ relative to the proximal segment's local frame as $\mathbf{Z} = \left[x_1, y_1, x_2, y_2\right]^T$. The cost function minimizes the squared Euclidean distance between the computed distal tip $(x_t, y_t)$ and the target $(x_d, y_d)$:

$$J(\mathbf{Z}) = (x_t - x_d)^2 + (y_t - y_d)^2. \tag{4}$$

The distal tip position $(x_t, y_t)$ is computed by sequentially transforming segment frames under the constant-curvature (CC) assumption:

$$\begin{bmatrix} x_t \\ y_t \end{bmatrix} = P_{xy}\left(\sum_{i=1}^{2} R_i \begin{bmatrix} x_i \\ y_i \\ l_i \end{bmatrix}\right), \tag{5}$$

where $P_{xy}$ projects onto the $xy$-plane. Each segment rotation $R_i$ is obtained from its bending angle $\theta_i$ and plane angle $\phi_i$ using the standard CC formulation based on Rodrigues' rotation [23]. The complete IK optimization problem with constraints is formulated as:

$$\mathbf{Z}^* = \arg\min_{\mathbf{Z}} J(\mathbf{Z}) \text{ s.t. } \begin{cases} \theta_{[1,2]} \leq \theta_{\max}, \\ \phi_2 = \phi_1 + \pi, \\ (x_s - x_d)(-x_d) + (y_s - y_d)(-y_d) > 0, \\ x_1 \cdot x_d + y_1 \cdot y_d > 0, \end{cases} \tag{6}$$

these constraints ensure physically feasible and safe manipulator motions. The first constraint limits each segment's bending angle to $\theta_{max}$. The second, third and fourth constraints define the S-shaped posture used in this study. Additionally, the second constraint sets a fixed bending-plane relationship between segments. The third constraint defines a reachable boundary, where $(x_s, y_s)$ is the neutral tip position if the distal segment were unbent (i.e. $(x_2, y_2) = 0$). The fourth constraint ensures that the proximal segment bends in the general direction of the target. This planar S-shaped configuration is commonly employed in single-port surgical manipulators to enhance triangulation and workspace coverage.

*2) Intersegmental Crosstalk:* In multi-segment soft manipulators, actuating a given segment $i$ can induce unintended motion in its neighboring segment $j$ $(i \neq j)$ due to mechanical coupling. To quantify this effect, we define the crosstalk displacement $\eta_{j,k}$ experienced by segment $j$ at control step $k$, resulting from actuation of segment $i$. Let $(x_j^{\text{pre}}, y_j^{\text{pre}})$ denote the tip position of segment $j$ in the unpressurized state, and $(x_j^{\text{post}}, y_j^{\text{post}})$ its position after applying pressure $q_{i,k}$ to segment $i$. The induced displacement is defined as:

$$\eta_{j,k} = (\Delta x_j, \Delta y_j)_k, \ \Delta x_j = (x_j^{\text{post}} - x_j^{\text{pre}}), \ \Delta y_j = (y_j^{\text{post}} - y_j^{\text{pre}}). \tag{7}$$

This disturbance $\eta_{j,k}$ is incorporated into each segment's task space as a localized translational offset, effectively capturing the geometric influence of intersegmental crosstalk as shown in Fig. 4(b). To model this coupling, we implement an LSTM network that predicts the induced displacement $\hat{\eta}_{j,k}$ for segment $j$ based on the recent actuation history of segment $i$:

$$\hat{\eta}_{j,k} = \mathcal{X}_j(q_{i,k}, q_{i,k-1}, \ldots, q_{i,k-m}), \quad i \neq j, \tag{8}$$

where $\mathcal{X}_j : \mathbb{R}^{3(m+1)} \to \mathbb{R}^2$ denotes the learned mapping from pressure history to the crosstalk-induced planar displacement. Note that the crosstalk model notation can be expressed interchangeably as $\mathcal{X}_j(q_i)$ or $\mathcal{X}_i(q_j)$, depending on which segment's displacement is being predicted based on the adjacent segment's actuation.

*3) Iterative Correction Process:* We introduce an iterative correction approach that refines segment-level commands by compensating for predicted intersegmental crosstalk. First, the optimization defined in (6) computes the desired segment tip configurations $\mathbf{Z}^*$, specifying both intermediate and distal segment positions in task space. These target points are then input into the inverse models $\mathcal{L}_i$ to predict corresponding actuator commands $\hat{q}_{i,k}$ as formulated in (3). Next, the predicted pressures are passed into the crosstalk model $\mathcal{X}_i$, defined in (8), to estimate the planar disturbances $\hat{\eta}_{i,k}$, which capture the effect of intersegmental coupling. To mitigate these disturbances, at each correction iteration $h$, the segment-level position is updated by subtracting the previously predicted crosstalk disturbance:

$$(x, y)_{i,k}^{(h)} = (x, y)_{i,k}^{(0)} - \hat{\eta}_{i,k}^{(h-1)}, \quad h = 1, 2, \ldots, H, \tag{9}$$

where $(x, y)_{i,k}^{(0)}$ denotes the initial target configuration, and $H$ is the total number of correction steps. Each updated position $(x, y)_{i,k}^{(h)}$ is reprocessed through the inverse models $\mathcal{L}_i$, followed by a new crosstalk prediction $\hat{\eta}_{i,k}^{(h)}$ via $\mathcal{X}_i$. This feedback loop continues until $H$ iterations are completed. The final corrected actuation commands $\hat{q}_{[1,2],k}^{H}$ are then deployed to the robot. In this work, we set $H = 4$ because command updates after the third iteration fell below 5%, indicating empirical convergence.

## V. Model Training

The manipulator is actuated by digital pressure regulators (Tecno Basic, Hoerbiger) controlled via a microcontroller (Arduino Uno Rev4) with a 12-bit digital-to-analog converter. The microcontroller communicates with Python scripts over a serial interface. Tip positions and orientations of each segment are tracked at 20 Hz using 5-DOF EM sensors (NDI).

### A. Data Collection Protocol

Data were collected under quasi-static conditions, allowing the system to settle for 2 seconds before each measurement. Pressure patterns were randomly generated by actuation 0, 1, or 2 chambers out of the 3 chambers per segment with probabilities 0.05, 0.25, and 0.7, respectively. Pressures for the selected chambers were randomly assigned within the safe operating range of 0 to 1.6 bar, as constrained by the robot's hardware. The resultant workspace reached was approximately 20 mm × 25 mm for the distal segment and 6 mm × 6 mm for the proximal segment. The robot was oriented horizontally to account for gravitational effects. Segment poses were recorded in a global reference frame $O$, with relative transformations computed as $T_i^{i-1} = \left(T_{i-1}^O\right)^{-1} T_i^O$, where $T_{i-1}^O$ and $T_i^O$ are the measured poses of segments $i-1$ and $i$ respectively.

*1) Single-Segment Data:* Each proximal and distal segment was actuated independently using 500 randomized pressure commands. These data were used to train the individual IK models (see Section IV-A).

*2) Crosstalk Data:* Crosstalk effects were extracted from the single-segment dataset by measuring the 2D displacement $\eta_j$ of segment $j$ when only its neighboring segment $i$ was actuated (i.e. $q_i \neq 0$, $q_j = 0$, with $j \in [1, 2]$). These displacement vectors and pressure inputs, totaling 500 samples, were used to train the crosstalk prediction model (see Section IV-B2).

*3) Two-Segment Data:* A two-segment dataset was constructed to train a multi-segment LSTM baseline for benchmarking the proposed RSCC approach. The single-segment data (Section V-A1) was merged with a multi-segment dataset comprising 1000 randomized actuation patterns with simultaneous pressurization of both segments. This resulting dataset, totaling 2000 samples, contains paired input states $[U_{1,k}, U_{2,k}]$. The multi-segment IK LSTM, denoted as $\mathcal{L}_{1,2}\colon \mathbb{R}^{16(m+1)} \to \mathbb{R}^6$, was then trained using this dataset to predict the actuating pressures for $q_{1,k}$ and $q_{2,k}$.

### B. Training Model Parameters

Datasets were divided by trajectory. Time-windowed samples from a given trajectory are assigned to train or validation (90:10), ensuring no overlapping windows cross splits. We performed a grid search on hyperparameters for **LSTM** (hidden size $\{64, 128, 256\}$, layers $\{2, 3\}$) and **Transformer** ($d_{\text{model}} \in \{128, 256\}$, layers $\{2, 4\}$, heads $\{4, 8\}$). Each configuration was evaluated at $m \in \{0, 2, 5\}$ using validation MSE (means ± SD over 5 seeds). All runs used Adam, batch size 32, dropout 0.15, learning rate $1 \times 10^{-4}$, and weight decay $1 \times 10^{-5}$. We selected the best-in-class models for subsequent experiments: LSTM = 2 layers, hidden size 128; Transformer = 2 layers, $d_{\text{model}} = 256$, 4 heads. The multi-segment LSTM baseline was configured with 3 layers and hidden size 128.

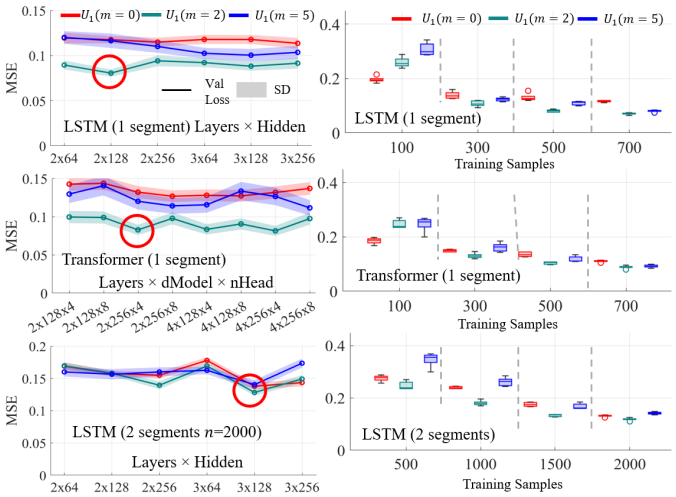


Fig. 5: Validation loss across different hyperparameters and time windows ($m$) for LSTM and Transformer models (left), and corresponding training-sample studies (right). Each point on the left represents the mean and standard deviation of validation loss across 5 random seeds. Red circles mark the best-performing configurations, which were selected for further evaluation with varying training samples in both one- and two-segment models.

Across 500 training samples, tuning hyperparameters reduced the validation loss by approximately 12–18% for both architectures, indicating moderate sensitivity to model capacity as shown in Fig. 5. However, changing the time window $m$ from 0 to 2 yielded a larger improvement of about 35–40%, confirming that temporal context contributes more to learning performance than network depth or width. For both LSTM and Transformer models, gains saturated when $m = 5$, suggesting diminishing returns for longer histories under limited data.

### C. RSCC Pipeline Benchmarking

The RSCC pipeline is not directly comparable to a single LSTM, as it integrates multiple segment-wise networks with iterative crosstalk subtraction. Nevertheless, we compare the learning behavior and data efficiency between individual LSTMs within RSCC and the monolithic two-segment model. As shown in Fig. 5, increasing training data per segment in RSCC yields a steeper decline in validation loss compared with a monolithic two-segment LSTM. Expanding the two-segment dataset from 500 to 2000 samples reduced validation loss by only 25–30%. Conversely, increasing single-segment data from 100 to 500 samples reduced it by 45–50%. This result suggests that the RSCC architecture leads to a more data-efficient learning pipeline. For single-segment models, both LSTM and Transformer improved with data, but at 700 samples, the Transformer with $m = 5$ achieved about 10–15% lower loss than the LSTM, suggesting an advantage when more data are available.

## VI. Experimental Validation

This section presents the experimental validation of the sequence-based models. Single-segment robot experiments evaluate the optimal time window $m$ prediction via model accuracy assessments. The time window length ($m$) that achieved the lowest RMSE in Fig. 7 was used for both the crosstalk model and LSTM model in two-segment experiments. The two-segment robot experiments evaluate the

TABLE III: Training and Validation MSEs for Different Time Window Lengths ($m$), Model Inputs, and sampling data ($n$)

| Model | Input | $m$ | Epochs | Train Loss | Val Loss |
|---|---|---|---|---|---|
| IK (Proximal, LSTM) | $X_1$ | 0 | 343 | 0.0674 | 0.1208 |
| IK (Proximal, LSTM) | $U_1$ | 0 | 379 | 0.0483 | 0.1059 |
| IK (Proximal, LSTM) | $U_1$ | 1 | 386 | 0.0358 | 0.0770 |
| IK (Proximal, LSTM) | $U_1$ | 2 | 384 | 0.0345 | 0.0744 |
| IK (Proximal, LSTM) | $U_1$ | 3 | 339 | 0.0348 | 0.0754 |
| IK (Proximal, Transformer) | $X_1$ | 0 | 48 | 0.0970 | 0.1120 |
| IK (Proximal, Transformer) | $U_1$ | 0 | 67 | 0.1035 | 0.1071 |
| IK (Proximal, Transformer) | $U_1$ | 1 | 56 | 0.1056 | 0.1068 |
| IK (Proximal, Transformer) | $U_1$ | 2 | 83 | 0.0797 | 0.1033 |
| IK (Proximal, Transformer) | $U_1$ | 3 | 78 | 0.0757 | 0.1015 |
| Crosstalk (Proximal $\rightarrow$ Distal, LSTM) | $q_1$ | 2 | 375 | 0.0380 | 0.0494 |
| Crosstalk (Distal $\rightarrow$ Proximal, LSTM) | $q_2$ | 2 | 382 | 0.0532 | 0.0819 |
| IK (Distal, LSTM) | $U_2$ | 2 | 388 | 0.0250 | 0.0509 |
| IK (Multi-Segment, LSTM, $n = 2000$) | $[U_1, U_2]$ | 2 | 396 | 0.0303 | 0.1199 |

performance of the RSCC pipeline in comparison to a multi-segment LSTM model, and in a load bearing condition. The models information was listed in Table III.

### A. Validation of a Single-Segment Robot

The single-segment study compared sequence models across time windows. We evaluated models with window lengths ($m \in \{0, .., 3\}$) using the augmented input $U_{i,k}$ from (3) and $m = 0$ using the input vector $X_{i,k}$. The models were evaluated using sequence Point-to-point, circular, and rectangular trajectories. To enlarge the reachable set during testing, we used distal tip sensing while actuating the proximal segment. Further details of these experiments are provided in the Supplementary Video.

*1) Point-to-Point:* 20 sequence points within the training workspace were used to generate seen inputs for the sequence-based models at various time windows. The distal tip position errors were recorded. The results are presented in Fig. 6. While all models performed comparably on the seen input sequences, the configuration with $m = 2$ achieved the lowest RMSE for both the LSTM and Transformer. This aligns with our findings in Section V-B. The Euclidean RMSE was calculated and reported in Table IV.

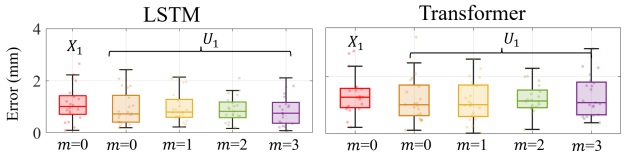


Fig. 6: Single-segment Point-to-Point results. Colored box plots indicate error bar between tip positions from predicted pressure and ground-truth tip for each model and input window($m$).

*2) Trajectory Control:* Predefined shapes, including circles with a radius of 8 mm with clockwise and anti-clockwise directions and a rectangular pattern with 10 mm width and 4 mm length (total 50 points), were generated with random origin in the x-y plane to evaluate model performance. These inputs were unseen by the training models. Additionally, comparisons with a geometric-based model derived from [2], data-driven models including Gaussian Process (GP) [24] and FNN [12] were conducted.

TABLE IV: Euclidean RMSE (mean [95% CI] in mm )

| | **P2P** $n = 20$ **points** | | **Trajectory** $n = 50$ **points** | | |
|---|---|---|---|---|---|
| $m$ | LSTM | Transformer | LSTM | Transformer | Models ($m$=0) [Ref.] |
| 0 ($X$) | 1.23 [0.77, 1.69] | 1.51 [1.02, 2.00] | 1.92 [1.30, 2.54] | 1.98 [1.39, 2.57] | 2.41 [1.68, 3.14] [2] |
| 0 ($U$) | 0.96 [0.40, 1.52] | 1.34 [0.84, 1.84] | 1.43 [1.00, 1.86] | 1.42 [0.98, 1.86] | 2.01 [1.51, 2.51] [24] |
| 1 ($U$) | 1.08 [0.68, 1.48] | 1.35 [0.77, 1.93] | 1.17 [0.72, 1.62] | 1.39 [0.95, 1.83] | 2.12 [1.32, 2.92] [12] |
| **2** ($U$) | **0.93 [0.62, 1.24]** | 1.37 [0.91, 1.83] | **0.80 [0.49, 1.19]** | 1.28 [0.86, 1.70] | |
| 3 ($U$) | 0.95 [0.52, 1.38] | 1.34 [0.89, 1.89] | 1.03 [0.68, 1.38] | 1.92 [1.24, 2.60] | |

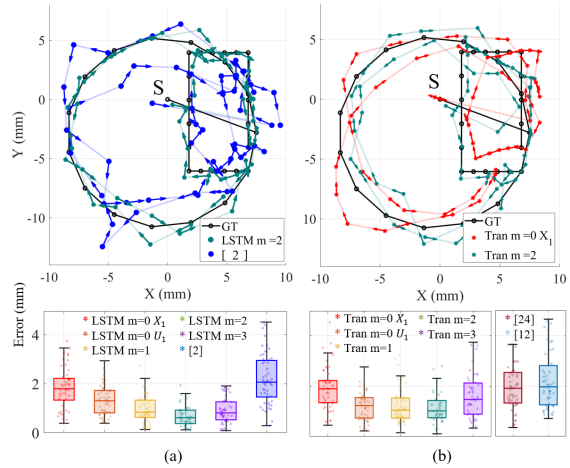


Fig. 7: Single-segment trajectory tracking on predefined circular and rectangular paths. Solid black lines denote ground-truth (GT) tip trajectories, while colored markers show robot motion driven by predicted pressure commands from different models. (a) LSTM-based comparison ($m = 2$) using input $U$, benchmarked against the geometric model [2]. (b) Transformer-based comparison ($m = 2$) with augmented inputs $U$, benchmarked against conventional input $X$. Boxplots indicate tip-position RMSE across trajectories with different time windows, Gaussian Process [24] and FNN [12].

The performance outcomes are shown in Fig. 7, and summarized quantitatively in Table IV. The LSTM at $m = 2$ again yielded the best performance, with the lowest RMSE among all models tested. This result confirms the superiority of the LSTM over the Transformer, GP, FNN and the geometric model-based approach.

*3) Effect of Sampling Density:* We evaluated the effect of sampling density on inverse kinematics accuracy using the best-performing model (LSTM, $m = 2$). The workspace was discretized into bins where point-to-point commands were given in a random sequence with each grid containing 3 commands. Local RMSE and the corresponding training sample counts were reported per bin. As shown in Fig. 8, regions with higher sample density generally exhibit lower RMSE. A Pearson correlation analysis confirmed a negative correlation between the sample count per bin and the resulting RMSE ($r = -0.94, p < 3.08 \times 10^{-10}$). This demonstrates the model's effectiveness in leveraging local data coverage to improve prediction accuracy. While minor deviations from this trend exist (e.g., bin $X \in [-5, 0], Y \in [0, 5]$), these likely stem from sequence-dependent hysteresis effects from preceding commands rather than insufficient local data.

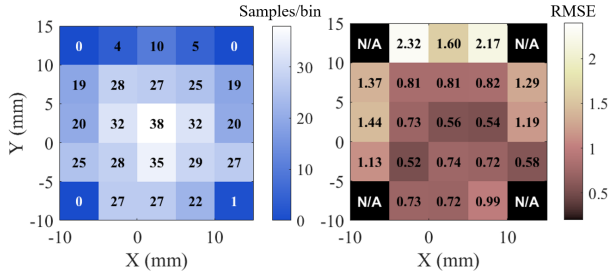


Fig. 8: Effect of sampling density on IK accuracy using the best LSTM model ($m = 2$). (Left) Sample count per 5 mm bin. (Right) Tip RMSE from three point-to-point tests per bin, showing lower errors in densely sampled regions but deviations due to hysteresis.

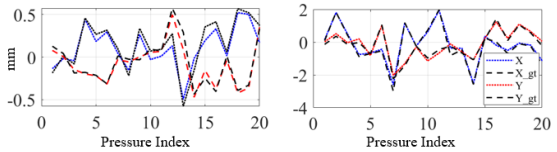


Fig. 9: Intersegment crosstalk displacement results. Predicted (solid line) vs. measured (dotted line) data across 20 randomized pressure commands. (Left) Proximal-segment disturbance (Right) Distal-segment disturbance.

### *B. Validation of the Crosstalk Prediction*

We validated the LSTM-based crosstalk prediction models described in Section IV-B2 using independent experimental data. For each segment, 50 randomized pressure commands were applied individually to the actuated segment $i$, while the adjacent segment $j$ remained unpressurized.

The resulting tip displacement offset $\eta_j$ was measured and compared with the predicted value $\hat{\eta}_j$ generated by model $\mathcal{X}_j$. The result is shown in Fig. 9. The RMSE of the Euclidean crosstalk offset was 0.230 mm (SD 0.128 mm) for proximal segment (from distal actuation) and 0.098 mm (SD 0.046 mm) for distal segment (from proximal actuation).

### *C. Validation of a Two-Segment Robot*

In two-segment experiments, we compared the performance of our proposed RSCC pipeline ($m = 2$) with a conventional LSTM model ($m = 2$) for a fair comparison on the hysteresis data. The commands for the proximal $U_1$ and distal segments $U_2$ were used as inputs to both the RSCC pipeline and the conventional LSTM model. We used a paired *t*-test to compare RMSE values from the RSCC and LSTM models on identical target points, testing the null hypothesis that their mean errors are equal. In this experiment, all data-driven controllers ran at 20 Hz. The RSCC pipeline required 28 ms per command, whereas the monolithic LSTM required 48 ms per command (tested on an RTX 4070, CUDA 12.8). Further detail in the Supplementary Video.

*1) Point-to-Point:* 50 random points were selected within the distal workspace as $(x_d, y_d)$. We calculated the command for each segment using the optimizer without imposing an S-shaped constraint.

The RSCC pipeline reduced proximal RMSE from 1.03 mm with LSTM to 0.70 mm , and distal RMSE from 2.68 mm to 1.20 mm, as shown in Table V ($p < 10^{-3}$) and Fig. 10.

*2) Trajectory Control with S-Shaped Posture:* To evaluate the effectiveness of the RSCC pipeline in handling different geometric constraints, we executed experiments involving circular trajectories $R$ as $(x_d, y_d)$, with 30 sample points per circle, while varying the proximal segment radius $r$ with defined $\theta_{1,\max}$ in (6).

The RSCC lowered proximal RMSE from 0.65 mm with LSTM to 0.47 mm, and distal RMSE from 1.62 mm to 1.10 mm ($p < 10^{-5}$). The RMSE and error bar of this experiment was reported in Table V and Fig. 11.

TABLE V: The Euclidean RMSE (mm) [95% CI] and p-value

| $i$ | **P2P** $n = 50$ points | | | **Circle S Shape** $n = 60$ points | | |
|---|---|---|---|---|---|---|
| | **RSCC** | LSTM | p | **RSCC** | LSTM | p |
| 1 | **0.40 [0.28, 0.52]** | 0.93 [0.37, 1.49] | $< 10^{-4}$ | **0.47 [0.28, 0.66]** | 0.65 [0.45, 0.85] | $< 10^{-5}$ |
| 2 | **1.20 [0.96, 1.44]** | 1.88 [0.76, 2.40] | $< 10^{-3}$ | **1.10 [0.86, 1.34]** | 1.62 [1.36, 1.88] | $< 10^{-5}$ |

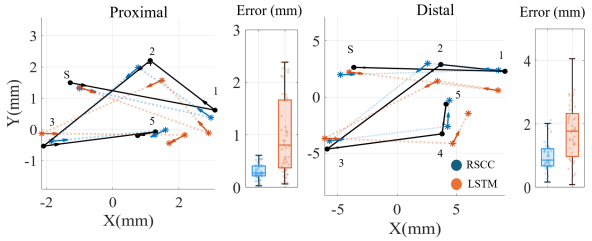


Fig. 10: Two-segment point-to-point example results (5 points) and Boxplots of Euclidean tracking error. (Left) Proximal segment tip positions (i = 1): black circles denote ground-truth; orange lines show tip coordinates from the conventional LSTM; blue lines show tip coordinates from RSCC. (Right) Distal segment tip positions (i = 2).

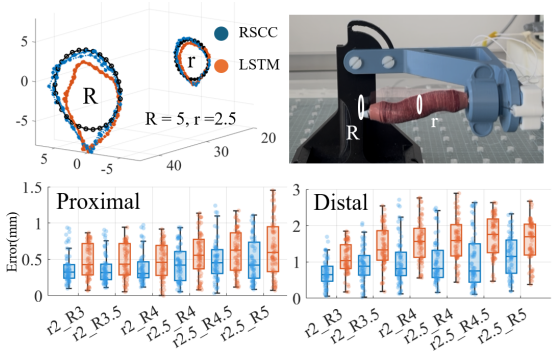


Fig. 11: Two-segment circular-trajectory results. Top: S-shaped posture—proximal segment traces a circle of radius $r$ (white), distal traces radius $R$. Left: ground-truth paths (black) with trajectories from RSCC (blue) and LSTM (orange). Bottom: boxplots of Euclidean tracking error for each $r$–$R$ pair (left: proximal; right: distal).

*3) S-Shaped Trajectory Under Payload (RSCC):* The robustness of RSCC under external loading was evaluated by tracking a circular distal trajectory while maintaining the S-shaped configuration ($R = 4.5$, $r = 2.0$, $2.5$ mm). Discrete tip loads of 5, 10, and 20 g were attached to emulate tissue traction. An additional 10 g trial was conducted with distal tendon actuation to mimic traction-assisted ESD, using identical RSCC pressure commands. Each test comprised 30 sample points per circle. Figure 12 shows the distal tracking performance under varying payloads. The median RMSE rose from about 1.0 mm (no load) to 1.1 mm (5 g), 1.6 mm (10 g), and 6.2 mm (20 g), while the 10 g tendon-actuated case achieved 2.2 mm. The high apparent error at 20 g mainly resulted from clamp displacement rather than tracking deviation, as trajectories remained geometrically consistent.

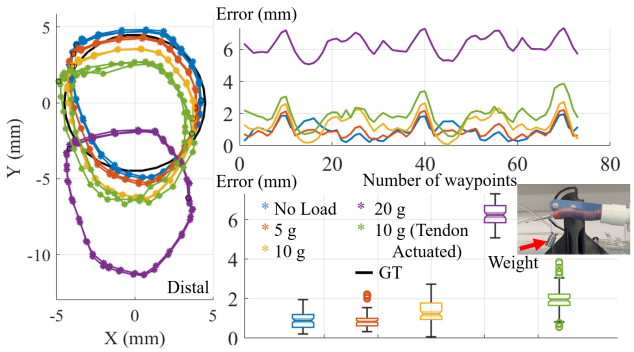


Fig. 12: Tracking performance of the distal segment under different payloads. Circular trajectories ($R = 4.5$ mm, $r = 2.5$ mm) were executed with 0–20 g tip loads and a 10 g tendon-actuated case.

## VII. Discussion & Conclusion

This work presented RSCC, a scalable inverse-kinematics framework for multi-segment soft continuum manipulators with morphable chambers. By combining sequence-based learning with recursive crosstalk compensation, the system achieves data-efficient modeling of hysteresis and intersegmental coupling, requiring only 500 training samples per segment collected under quasi-static conditions. Because our evaluation focuses on bending-dominant behavior in a compact workspace, absolute sample counts are not compared across studies; instead, we report within-system learning curves and workspace-coverage results, as shown in Fig. 8.

The proposed soft manipulator exhibits notable hysteresis with task-space path dependence (see Section III-A). Multi-variable Bouc–Wen or Preisach formulations, which model hysteresis in the input domain, would require augmentation to account for task-space trajectory history. Such an increase in the dimensionality of models makes parameter identification intractable [10]. In contrast, data-driven models such as the LSTM and Transformer implicitly capture these effects without explicit parameter tuning, providing a scalable solution for complex multi-segment soft robots.

In the single-segment case, augmenting the target state $U$ with historical information improved hysteresis modeling compared to conventional task-space input $X$, reducing RMSE on unseen data from 1.92 mm to 1.43 mm ($p < 10^{-3}$). The optimal time window ($m = 2$) with limited data balanced accuracy and generalization, improving circular trajectory tracking RMSE from 1.43 mm to 0.8 mm. Shorter windows underfit hysteresis, while longer ones risked overfitting. In contrast to previous sequence-based studies [13], [14], [20], which used substantially larger training datasets and fixed or arbitrary window sizes, the present results indicate that careful time-window selection becomes far more influential when data are scarce. Across the explored hyperparameter ranges and time windows, LSTMs consistently outperformed Transformers for $\leq 500$ samples per segment, with Transformers reaching comparable performance only when data increased to 700 samples; this suggests that, in data-limited soft robotics scenarios, simpler recurrent architectures may be preferable to higher-capacity Transformers. The final LSTM model achieved approximately 6% RMSE relative to the command workspace, outperforming the 13%-15% RMSE for the model-based and non-parametric baselines [2], [12], [24].

The RSCC pipeline demonstrated superior data efficiency compared to a monolithic two-segment LSTM. RSCC achieved precise tip tracking with only one-quarter of the data required by a conventional two-segment model. Distal and proximal segment RMSEs reached 11 % and 7.8 % of their commanded workspaces (3.4 % and 5.5 % relative to the training workspace). Note that the command workspace was intentionally constrained to maintain the robot within the S-shaped posture. The framework remained robust under loading: distal error stayed below 3 mm for payloads up to 10 g, both with and without tendon actuation, while the apparent 6.7 mm deviation at 20 g was largely due to clamp displacement rather than control drift.

In surgical robotic telemanipulation contexts, minimizing sensor integration is essential for robustness. Thus, achieving clinically acceptable open-loop accuracy is more practical than relying on sensor-intensive closed-loop control, as the human remains in the loop throughout the surgical procedure. Fine positional adjustments are managed by tendon-driven actuation under endoscopic visualization. Future work will improve robustness and applicability via better data collection. We will add input normalization and three-chamber pressurization to increase lateral stiffness and payload capacity for ESD, and validate the controller ex vivo for segments outside the direct visual field.

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